

William Hao-Cheng Huang

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Education

University of California, Berkeley

California, USA

M.A., Applied Statistics (GPA: 3.72/4.00)

Aug. 2020 - May 2021

- Coursework: Probability Theory, Mathematical Statistics, Linear Model, Statistical Computing, Applied ML

National Chiao Tung University

Taiwan

B.S., Applied Mathematics (GPA: 4.05/4.30 (Top 5%))

Sep. 2015 - July 2019

B.S., Undergraduate Honors Program of Interdisciplinary Science, College of Science

- Phi Tau Phi (Top 1% in the class), College of Science Dean Award
- Coursework: Real Analysis (HKUST), Introduction to Analysis I–II, Probability Theory, Advanced Statistics, Large Sparse Matrix Computations (grad.), Scientific Computing (grad.), Algorithms (grad.)

Research Outputs

Huang, W. H.-C. *Spectral Thresholds for Identifiability and Stability: Finite-Sample Phase Transitions in High-Dimensional Learning*. arXiv preprint:[2510.03809](https://arxiv.org/abs/2510.03809) (2025).

- Studies a finite-sample Fisher-spectral cutoff predicting when high-dimensional inference becomes unstable.

Huang, W. H.-C. *Spectral Identifiability for Interpretable Probe Geometry*. arXiv preprint: [2511.16288](https://arxiv.org/abs/2511.16288) (2025).

- Investigates an eigenspace-reliability criterion that tests probe reproducibility through Fisher-spectral stability.

Huang, W. H.-C. (**Co-Inventor**). *Method for Critical Dimension Recovery by H₂/He Plasma Treatment in EUV Mask Process*. Patent applications filed in the U.S., Netherlands, China, Japan, and Taiwan, 2025. (*pending*)

- Develops an inverse-estimation method for nanoscale recovery formulated as a model-guided correction process.

Huang, W. H.-C. *A Graph-Theoretic Approach for Protein Simulation*. Presented at the Taiwan Society for Industrial and Applied Mathematics (SIAM) Annual Conference, 2018. (**Most Excellent Poster Paper**)

- Builds a Hessian probabilistic model for posterior vibration inference with graph-structured numerical priors.

Applied Research Experience

Taiwan Semiconductor Manufacturing Company (TSMC)

Taiwan

Research Engineer

Feb. 2022 - Oct. 2025

- Developed a Fisher-spectral identifiability method to characterize stability in high-dimensional probe inference.
- Derived a spectral reliability criterion for principled geometric decomposition of process-driven variation.
- Proposed a likelihood-based comparison method for heterogeneous conditions with controlled error rate.
- Designed a counterfactual model cutting Type-I error from 7% to 3%, avoiding about USD 100K weekly loss.

National Center for Theoretical Sciences

Taiwan

Research Assistant

Aug. 2019 - July 2020

- Built a hierarchical Bayesian model for rare-event estimation with hospital-level group effects and latent terms.
- Quantified uncertainty under urban-rural heterogeneity using structured shrinkage priors, achieving AUC 0.95.
- Developed a posterior-predictive framework that guided interventions benefiting $\approx 1,600$ patients annually.

The University of Tokyo, Graduate School of Mathematical Sciences

Japan

Researcher Visitor

Jan. 2019 - Mar. 2019

- Investigated Jacobi-Davidson and projection methods for large-scale eigenvalue problems in dominant systems.
- Analyzed correction equations, preconditioning schemes, and harmonic Ritz methods for improved convergence.
- Applied the resulting numerical methods to spectral option-pricing models for stable eigenstructure estimation.

National Chiao Tung University, Applied Mathematics

Taiwan

Undergraduate Researcher

Sep. 2018 - July 2019

- Studied ill-conditioned configurations in acoustic localization and identified causes of estimation failure.
- Derived a noise-geometry sensitivity criterion delineating stable and unstable regions in inverse problems.
- Validated the approach on indoor localization experiments, showing improved stability under noise.

Competitive Positions and Awards

Industrial Research Recognition, Taiwan Semiconductor Manufacturing Company 2022 - 2025

- Entrusted with ten confidential modeling projects formally registered as proprietary industrial research.
- Several results later informed ongoing academic work on mathematical stability and reliability.

Phi Tau Phi Award 2019

- Awarded for top academic standing in the College of Science.

College of Science Dream-Building Scholarship Program 2019

- Awarded USD 2,500 merit scholarship for outstanding academic performance.

Outstanding Achievement in Mathematics Coursework 2016 - 2019

- Recognized multiple times for excellence in undergraduate mathematics coursework.

Outstanding Achievement in Coursework 2016 - 2019

- Cited repeatedly for highest academic performance in the Applied Mathematics program.

The Most Excellent Poster Paper in Taiwan Society for Industrial and Applied Mathematics 2018

- Developed a Hessian-based probabilistic simulator that achieved a 66% acceleration over baseline methods.
- Received Most Excellent Poster Award at the 6th Annual Conference of SIAM Taiwan.

Research Recognition, Institute of Mathematics, Academia Sinica 2018

- Conducted a project integrating computational methods and inference for signal-processing tasks.

Leadership and Teamwork

Taiwan Semiconductor Manufacturing Company (TSMC) Taiwan

Industrial Research Project Lead Feb. 2024 - Oct. 2025

- Led a six-person research team and mentored peers in probabilistic programming for modeling applications.
- Conducted a mathematical workshop for ~50 managers and two technical sessions for ~200 engineers.
- Coordinated project milestones and communication between data science, modeling, and production units.

University Infinite Research Club of Mathematics and Science Taiwan

President Aug. 2016 - Jan. 2019

- Expanded membership from 3 to 40 and established a leading student research community in Applied Math.
- Organized 4 panels and 32 case-study sessions with 30+ participants, fostering academic collaboration.
- Mentored 10 undergraduates, helping them complete substantive research projects.

Academic Affairs in Applied Mathematics Student Association Taiwan

Director Sep. 2016 - July 2018

- Coordinated 4 workshops, 15 alumni talks, and 16 information sessions with 70+ attendees.
- Edited and published the Applied Mathematics departmental newsletter for academic outreach.
- Provided course-selection guidance for 100+ freshmen in the Applied Mathematics program.

Professional Service

National Chiao Tung University, Program of Interdisciplinary Science Taiwan

Teacher Assistant Mar. 2018 - Jan. 2019

- Led technical discussion sessions for 50 students in interdisciplinary science courses.
- Designed interdisciplinary course modules and hands-on workshops across two semesters.
- Advised students on interdisciplinary research projects through scheduled project consultations.

Skills

- **Research Expertise:** probabilistic modeling, spectral analysis, optimization, numerical and bayesian inference
- **Programming:** Python, C++, MATLAB, R, SQL, Fortran, Bash, high-performance and probabilistic computing.
- **Tools:** Linux, TensorFlow, PyTorch, Scikit-learn, Pandas, NumPy, SciPy, Spark, SAS, LaTeX, Git